Seismology

An open instrument for monitoring — and the lesson about what you subtract before you count

Level: Explorer and Scientist — ages 11–19 (Romanian school years V–XII)
- **Time in class:** 20 min demonstration, 50 min investigation, or a 2–3 week project
- **Sensors:** MPU-6500 (acceleration and angular rate)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer and Scientist — ages 11–19 (Romanian school years V–XII). As a demonstration it works from year V. As a *monitoring instrument* — for a project that records for days — it needs patience and a log, so from year VII. The most direct entries are in geography, **year V** (earthquakes) and **year IX** (seismicity), and in **years IX–XII** (research method, monitoring projects).

Assumed knowledge

The guided activity *The Seismometer*, done first: the silence protocol, the noise floor and the distinction between magnitude and intensity all come from there. In addition: Pythagoras in three dimensions, if you want to explain the formula for the length of a vector.

Time in class

Short: 20 min (demonstration). Full: 50 min (one investigation). Project: 2–3 weeks of monitoring at 10 min a day, plus two analysis lessons.

Teacher preparation

20 min — choose and **test** the monitoring spot; mark the box's position with tape; check the WiFi coverage there; and run the app for five minutes to see **what the event counter does** on an ordinary day.

Sensor settling time

Ten seconds of silence before every deliberate reading. For monitoring, the box stands still for days — **and moving it, even by a few centimetres, breaks the series**.

Sensors and boxes

MPU-6500. One box is enough; two allow two places in the building to be compared, which is the best investigation here.

Three routes

Short	20 min: projected demonstration — the five cards, the seismograph line, and the test that shows *what* the event counter counts: shakes, not seconds. It ends with the **Theory** tab: magnitude, intensity and the 1977 earthquake.
Full	50 min: one investigation — comparing two places in the building, or hunting the source of the school's background noise over an hour.
Project	2–3 weeks: monitoring at a fixed spot with a paper log, and comparing school days with the weekend.

The step-by-step activity first, then the open instrument

***The Seismometer* is done first.** The things without which this app means nothing come from there: the silence protocol, the idea of a noise floor, and the distinction between magnitude and intensity.

And there is a better reason. The two resources *show* different things, and comparing them is the best lesson in the pair. The guided activity **subtracts gravity** with a filter and draws small departures on a ±0.25 g scale. The app **keeps it** on the **Shaking** card, in g, so at rest that card reads about 1 g rather than zero — which is the honest raw reading. **But on the seismograph line, and before deciding what counts as an event, it subtracts gravity too**, exactly as the exercise does, because otherwise the direction of a shake would matter more than its size.

A class that has seen both can answer a question neither asks on its own: *what exactly did you subtract, and why does it matter?*

1. What the app shows and what it does not

It is an **open instrument**: no steps, no green button, no observation journal. **You set the pace.**

- **Five cards**: **Shaking**, in g, **Roll** and **Pitch** in degrees, and **Accel. X** and **Accel. Z**, in g.
- **The seismograph line** — the chart **Shaking away from rest**, showing the last 30 seconds —, which draws the departure from rest, gravity subtracted, so in silence it sits near zero. The event threshold is drawn as a **dashed red line** labelled **event threshold 0.2 g**, and the start of every event is marked with a red vertical line. The vertical axis starts at zero and shows at least 0.5 g; a bigger shake widens it.
- **The event counter**, at the top next to the title, starting at **0 events detected**.
- **Three tabs**: **Accelerometer** (the three acceleration axes), **Gyroscope** (the three rotation axes) and **Theory** — what a seismograph is, magnitude and intensity, the Vrancea earthquakes. On a small screen the tabs scroll: three stacked charts do not fit under the seismograph at 1366 × 768, so scroll down to reach the third.
- **The Export data button**, top right.

Technical note — the Shaking card, the threshold, and what is subtracted before counting

The sensor. MPU-6500. **The motion sensor is mounted upside down inside the box, and the firmware already corrects for it**: a box lying flat reports $AZ \approx +1$ g. A box reading −1 g flat is either running old firmware and needs reflashing, or has its sensor fitted the other way up: tick **Motion sensor upside down** for it under **Settings → Box names**.

The app reads **every sample** from the box (50 a second), so a short knock does not slip between two readings. From firmware 2.2.5, the chip smooths the accelerations with a filter of about 20 Hz: footsteps, doors and knocks, all below about 25 Hz, are reproduced faithfully, and the hum of motors no longer shows up as false trembling.

The Shaking card includes gravity. It shows, in g,

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2},$$

and gravity never switches off: **at rest the card reads about** 1 **g, not zero.** The card used to be called *Magnitude*, and pupils read "1.01" as an earthquake's magnitude; the new name says what it is: how hard *the box* is being shaken, in g. **It is not a seismic magnitude** and is not compared with the numbers in the news.
The threshold is 0.2 **g — but from what.** Not from zero, and not from a fixed 1 g: the app **learns for itself where "still" is**, as a three-component vector, and compares every reading against that. An event begins when the departure from rest passes 0.2 g — that is, when the seismograph line crosses the dashed red line.
Why it is subtracted as a vector and not as a number. Subtracting from the *length of the vector* would leave direction mattering: a horizontal shake enters the length squared, so it would almost vanish. Subtracting the resting vector makes **a shake of the same size count the same from any direction** — something a class can check in thirty seconds, and which the theory box below explains.
The counter counts shakes, not tenths of a second. A shake is counted **once**, as it starts, however long it lasts: two seconds of vibration add 1, not twenty. To be counted again the signal must drop below 0.12 g and stay quiet for about three tenths of a second — which is why a shake that wobbles around the threshold is still **one** event.
What still needs saying to a class. The number on screen is a count of *shakes the box felt* — footsteps, doors, a chair pushed back — not a count of earthquakes. That distinction survives in full, and it is the lesson: an honest instrument counts correctly, but you still have to know what it counted.
Roll and pitch do not depend on the sign of AZ**.** The formulas use $\sqrt{a_x^2 + a_z^2}$ and $\sqrt{a_y^2 + a_z^2}$, so a_z enters squared. Worth knowing when comparing with other activities.
Exporting. The **Export data** button, top right. **The IMU file is separate** from the slow-sensor one — look for the `_imu` suffix — because it is sampled at a different rate. **The event counter is not exported**: it lives only on screen and is lost when the app closes, so it is written down on paper.

2. Where this fits in the curriculum

- **Geography** — in **year V**, earthquakes; in **year IX**, seismicity and the Earth's internal structure. Multi-day monitoring gives what a single lesson cannot: a *time series* of their own.
- **Physics** — in **year VII**, the addition of vectors: the length of the acceleration vector is Pythagoras in three dimensions, and the *reason* the app subtracts gravity as a vector rather than as a number is a direct consequence of vector addition. **It is the best application of vectors in the whole collection**, because the screen shows what happens when you do it right and the arithmetic shows what would happen if you did not.
- **Mathematics** — in **years VII–VIII**, square roots and Pythagoras; means and distributions, if you analyse the multi-day series.
- **Computing and ICT** — processing an IMU file with many thousands of rows: filtering, a windowed maximum, a chart. It is a first real encounter with a large dataset.
- **Emergency preparedness** — the bridge to the *Earthquake!* activity.

The core skill is **reading an aggregate indicator critically**. The event counter shows a large and impressive number. Asking *what it actually counts* is exactly the skill the activity teaches.

3. Time plan

Route	Time	What happens
Demonstration	**20 min**	The screen projected. The five cards, the Shaking card sitting at 1 g at rest, then **the direction test**: a vertical shake and a horizontal one of the same strength trip the counter alike. It ends with the question: what does the counter count?
Full	**50 min**	One investigation: comparing two places in the building — ground floor and upper floor, or beside a load-bearing wall and in the middle of the room — ten minutes each, with a log. Or following a whole lesson and matching the peaks to the school timetable.
Project	**2–3 wks**	Monitoring at a fixed spot with a paper log, comparing school days with the weekend — when the building is empty and the background noise visibly falls.

4. Materials

Item	Per class	Notes
MagicBox units with MPU-6500	1–2	One is enough; two allow two places to be compared simultaneously, which is the best investigation.
Laptop with the app	1	For two boxes, two windows are opened.
A fixed, stable spot	1–2	Directly on the floor, beside a load-bearing wall, or on a heavy table. **Marked with tape**, because moving it breaks the series.
Mains power or a power bank	1–2	Multi-day monitoring needs continuous power.
Paper log	1	The time, what was happening in the building, whether it was break time, whether there were works. **Without it, a week of peaks is uninterpretable.**
Marking tape	1 roll	The box's position and, for the direction test, the shaking directions.
Shared folder	1	One folder per day, because the file names repeat.

Preparation. Choose the spot and *test* it: run the app for five minutes during an ordinary lesson and watch the counter. **If it gathers hundreds of "events" in one break**, the spot is too close to a corridor or a door and should be moved — or, on the contrary, that is exactly the subject of the project. Mark the position with tape and check the WiFi there.

5. Scientific background for the teacher

The length of a vector, and why gravity is subtracted as a vector

The accelerometer gives three numbers. The app combines them into one, by Pythagoras in three dimensions:

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2}.$$

The advantage is that a shake counts whichever direction it comes from — no axis has to be chosen as "the one that matters".

The disadvantage, which is not obvious, comes from the fact that **gravity is in the sum too**. At rest $a_z \approx 1$ g and the other two are nearly zero, so $|a| \approx 1$ g.
Now add a small shake. **If it is vertical**, it adds directly: $|a| \approx 1 + \delta$. **If it is horizontal**, it adds as a vector:

$$|a| = \sqrt{1 + \delta^2} \approx 1 + \frac{\delta^2}{2},$$

so the effect is *of order* δ^2, not δ. For $\delta = 0.2$ g, the vertical case gives 1.20 g and the horizontal one 1.02 g. **The same shake, two very different numbers**, purely because of direction.
This is exactly the problem a real seismograph solves by subtracting gravity *before* anything else — and which this app solves too, as does the guided activity with its filter.
And notice what has to be subtracted. Taking 1 g off the length of the vector is not enough: the arithmetic above shows that the horizontal shake has already entered squared, and no later subtraction gets it back out again. What must be subtracted is the resting **vector**, component by component, and only then is the length of what remains taken. A 0.2 g shake then gives 0.2 g whichever way it points.
It is the best vector lesson the platform can offer, because it is not theoretical: the class can check on the counter that direction no longer matters, and work out on paper how much it would have mattered had the subtraction been done wrongly.

5.1 Magnitude, intensity and the Vrancea earthquakes — the Theory tab

The app's **Theory** tab says the same as *The Seismometer* exercise, and is worth reading with the class at the end.

Magnitude describes the earthquake at its source: how much energy it released. Today the **moment magnitude**, M_w, is used; the "Richter scale" (local magnitude, M_L) is its historical ancestor. One unit up means ground motion ten times larger and about 32 times more energy. **Intensity** describes what was felt in one particular place; in Romania and the rest of Europe it is measured on the **European Macroseismic Scale, EMS-98**, in twelve degrees (from I, not felt, to XII, completely devastating). An earthquake has one magnitude and many intensities. The box measures a quantity of the intensity kind: the shaking where it stands.

Vrancea is a special case. Its large earthquakes happen at **intermediate depth, between about** 60 **and** 200 **km**, not near the surface. That is why they are felt over a very wide area, far into Moldova, Bulgaria and Ukraine, and why Bucharest, about 150 km away, is so exposed. The earthquake of **4 March 1977** was M_w 7.4, at about 95 km depth, and caused most of its casualties in Bucharest.

Early warning. Seismic waves do not all arrive together: the fast, relatively gentle P waves arrive first, and the S waves that cause the strong shaking come after them. Stations in Vrancea detect the earthquake at once, and the electronic signal reaches Bucharest almost instantly. Between that signal and the arrival of the strong waves there remain **about** 25 **seconds**. It is not much, but it is enough to stop a train, shut off a factory's gas or take cover — exactly what the *Earthquake!* activity practises.

5.2 What the event counter counts

A shake is counted **once, as it begins**. To be counted again the signal must fall below a level lower than the trigger and stay quiet for about three tenths of a second.

The practical consequence, and the test to do in class: a single two-second shake adds **one**; five short, distinct knocks add **five**. The two situations are entirely different and the counter tells them apart — which means a large number really does mean many shakes, rather than one long one.

What still has to be interpreted. The counter counts what the box felt, not what happened in the Earth's crust: footsteps in the corridor, a door slammed, a chair pushed back. The valid comparison is between intervals of the same length — "34 shakes over the long break, 2 during the mathematics lesson" — and the sentence that still cannot be written is "we detected 34 earthquakes".

Checking it against the exported file. From the IMU data the groups in which the departure exceeds the threshold can be counted independently, requiring a stated gap between them, and the result compared with what the app displayed. It is a spreadsheet exercise within reach at upper-secondary level, it is the activity's computing task, and it now has an expected answer: the two numbers should agree closely. **Checking an instrument that works** is a better habit than working around one that does not.

5.3 A building's background noise

A school building is never quiet during the day: footsteps in corridors, doors, furniture being moved, traffic outside. This is **anthropogenic background noise**, and it has a very regular daily pattern — it rises at arrival, falls during lessons, explodes at break time, disappears in the evening.

Comparing a school day with a weekend is the most convincing measurement in the project, because it shows a pattern everybody can predict and nobody has seen measured. And it explains, better than any text, why seismologists site their stations far from cities and why urban ones work mainly at night.

6. Ready-made investigations

6.1 Investigation 1: the direction test — first, always

Research question. Does the counter respond equally to vertical and horizontal shaking?

Method. The box on the table. Record the counter. Give **five vertical knocks** of moderate strength — a palm on the table beside the box — and record the counter. Wait for silence. Then **five horizontal shakes** of the same strength — a sideways push of the table — and record again.

What to measure. The rise in the counter for each set of five.

Expected finding. The two sets raise the counter by about the same amount — and that is a result, not the absence of one. The instrument was built not to tell directions apart, by subtracting the resting vector before comparing.

The physics comes straight afterwards, on paper: *how much would direction have mattered had the subtraction not been done?* Using the formula in the theory box, pupils work out that a 0.2 g shake would have given 1.20 g vertically and only 1.02 g horizontally — crossing a raw 1.2 g threshold in one case and not the other. **The investigation measures an absence, and the arithmetic shows how large the presence would have been.**

Do it before any comparison between places anyway, as a check on the rig: if your two sets come out very differently, something is wrong with the setup — the box sliding, the table rocking — and the comparisons that follow would not be valid.

6.2 Investigation 2: the counter counts shakes, not seconds

Research question. What does the counter add up?

Method. Record the counter. Make **one long shake**, about two seconds, keeping the table vibrating. Record. Then, after silence, **five short, distinct knocks**. Record again.

What to measure. The rise in the counter in each case.

Expected finding. The long shake adds one, and the five distinct knocks add **five**. The conclusion is written exactly so: *the counter counts shakes, not how long the shaking lasted.* If the knocks come too close together, with no quiet between them, two can merge into a single event — worth noticing too.

6.3 Investigation 3: two places in the building

Research question. Does every place in the building shake the same?

Method. Two boxes, simultaneously, ten minutes, in contrasting places: ground floor and upper floor, beside a load-bearing wall and in the middle of a room, or by the corridor and by the window. **The same height and the same orientation** for both.

What to measure. The rise in each counter, and the peak value of the seismograph line.

Expected finding. Upper floors move **more** than the ground floor, and the middle of a slab more than the edge by a load-bearing wall — exactly what building engineering says. Differences of two or three times are common.

6.4 Investigation 4: a school day against a weekend

Research question. How much of the building's agitation do we produce?

Method. The same box, the same spot, continuously powered. Record a full school day and, separately, a weekend day. Note the counter every hour, or export and process.

What to measure. The counter's hourly rise, for each day.

Expected finding. A very clear daily pattern on the school day, with peaks at break times, and a nearly flat weekend. **The ratio between them is the answer**, and it is usually tenfold or more.

The investigation, in one table

The question. What does the event counter actually count?
The hypothesis. Written **beforehand**: a two-second shake will add _____ to the counter, and five short knocks will add _____. Not changed afterwards.
What I change. One thing only: the **duration** of the shaking — long once, short and repeated the second time — at the same strength.
What I measure. The rise in the counter, written on paper before and after each set. The counter is not exported and is lost when the app closes.
What I keep the same. The same box, the same place, the same strength of knock, **the same direction** — so as not to change two things at once; investigation 1 checks separately that direction does not matter — and silence between sets.
How many repeats. Each set is done twice. If the two repeats differ greatly, the strength of the knocks was not constant.
What makes a reading valid — and what makes it an artefact. Valid is a reading with the counter noted immediately before and immediately after, with nobody else touching the table in between. An artefact is anything else: somebody walking past adds to the counter without knowing; a knock so hard that it holds the threshold exceeded for whole seconds inflates the counter by itself; and **a comparison between a vertical and a horizontal shake is not a comparison of strength** but of direction.

7. Analysis after the activity

7.1 What the data should look like

Situation	Typical value	What it means
Box at rest, the Shaking card	0.98–1.02 g	**Gravity.** Not zero, and not a magnitude.
The seismograph line, at rest	about 0 g	Gravity subtracted; what is left is noise, below a few hundredths of a g.
Event threshold	0.2 g from rest	Measured after the resting vector is subtracted, so direction does not enter.
Five moderate vertical palms	counter +5	One per knock.
Five horizontal pushes of the same strength	counter +5	**Direction does not matter — the result of test 1.**
One 2 s shake	counter +1	One shake, however long it lasts.
A quiet lesson	counter +0 to +3	Background noise.
The long break	counter +20 to +80	The building full of footsteps.
Weekend, the same hour	counter under +3	**The ratio is the project's result.**

7.2 Questions to ask afterwards

1. The Shaking card reads 1.01 g when the box is standing still. Why, and why did *The Seismometer* read zero in the same situation? Was it a magnitude-1 earthquake?
2. A palm on the table and a sideways push of the same strength raised the counter alike. Why was that not a foregone conclusion? Explain using the addition of vectors: what would have happened had the app taken 1 g off the length of the vector?
3. The counter rose by 34 over the long break. How many earthquakes were there? What did that number actually count?
4. The 1977 earthquake was M_w 7.4. How many times weaker would the motion of a 6.4 have been? Why does Bucharest get about 25 seconds of warning?
5. How would you count *distinct events* from the exported file? Why do you need an extra rule besides the threshold?
6. The upper floor moved more than the ground floor for the same sources. What does that mean for somebody choosing where to put a sensitive instrument?

7.3 Assessment ideas

- **Low stakes:** the pupil records the counter's rise for the two directions and says whether direction mattered.
- **Standard:** the pupil shows, with investigation 2, that a long shake counts once, and explains through vector addition why direction no longer matters.
- **Extension:** the pupil counts distinct events from the exported file, with a stated separation rule, and compares the result with the app's counter — explaining why the two should agree and what it would mean if they did not.

Common pitfalls and misconceptions

Reporting the counter as a number of earthquakes. It counts shakes the box felt — footsteps, doors, chairs. Do not write "we detected 34 earthquakes".
Comparing intervals of different lengths. A count of shakes grows with the time spent, so a 20-minute break does not compare with a 50-minute lesson. Compare equal intervals, or divide by the duration.
Expecting Shaking to be zero at rest. The card includes gravity, so it reads about 1 g. It is not a magnitude, and it is not broken.
Moving the box during a multi-day series. Even a few centimetres, or a different orientation, changes everything. Mark the position.
Forgetting to note the counter. It is not exported and is lost when the app closes. Write it on paper, before and after every interval.
Neglecting the paper log. A week of peaks with no notes about what was happening in the building cannot be interpreted at all.
Confusing the IMU file with the ordinary one. The acceleration data are in a separate file, with the `_imu` suffix, and have far more rows.

Safety and classroom management

The box is not struck. The table or the floor is struck, never the instrument.
Heavy furniture is not shaken. A sideways push of a table is made gently and under control, by one person. Nobody climbs on a table, nobody tips one over, and cupboards and shelves are not used as vibration sources.
Equipment left recording for days is announced to the school office and to colleagues who use the room, labelled with the class and teacher, and does not stand in an escape route or anywhere it can be knocked.
Cables. Continuous power means a cable in place for days: it does not cross walkways and is taped down.
The conversation about earthquakes is conducted as a matter of preparedness rather than fear — see the note in *The Seismometer* guide and the *Earthquake!* activity.

Adaptations for pupils with special educational needs

The counter, not the graph. The main adaptation: the pupil works only with the *rise in the counter* — a single number, noted before and after — rather than with the shape of the seismogram. The conclusions of investigations 1, 2 and 4 are all reached that way, with simple subtractions.
The "before / after / difference" table, printed as such, so that the pupil does not have to remember what it was before.
Three real roles for pupils with motor difficulties. The activity involves knocking and walking, but the roles that matter are done from a desk:

- **the direction keeper** — checks before each set that the knocks are all in the same direction, and has the right to void a mixed set; this is the role with the greatest scientific value, because direction is the activity's hidden variable;
- **the counter reader** — says the value before and after;
- **the scribe** — writes it down and computes the difference.

Noise. The knocks are sudden. Each set is announced beforehand, which is part of the protocol anyway.

If things do not go to plan

The counter rises constantly with nobody doing anything. The spot is too busy — next to a corridor, a door or a street. Move the box, or **make that the subject**: where is the quietest place in the school?
The counter never rises, not even on knocks. The knocks are too weak: the threshold is 0.2 g, far above a finger tap. Try a firm palm right beside the box — and if it still does not move, check that the app is receiving data by watching the cards.
You have only one box. Investigations 1, 2 and 4 are done in full with one. Only the comparison of two places (investigation 3) needs two — and it can be done sequentially, ten minutes at each spot, if you record the times.
You cannot leave the box overnight. Compare two hours of the same day — a break and a lesson — instead of two days. The pattern is the same, on a smaller scale.
The box will not connect. Do not stop the lesson. Use the "What the data should look like" table as a printed dataset: pupils compare the counter's rises by direction and by duration and draw the two conclusions — direction does not matter, and a long shake counts once. **The whole reasoning part — which is the lesson — is done without the equipment.** Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

8. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **Romanian National Institute for Earth Physics** — `https://www.infp.ro` — Romanian earthquakes in real time and the historical catalogue; the most relevant source for a class here.
- **IRIS / EarthScope** — `https://www.iris.edu` — material on school seismographs and how to read a seismogram, plus networks of educational instruments.
- **USGS Earthquake Hazards** — `https://www.usgs.gov` — search for seismic background noise and its daily patterns; it explains exactly the pattern the project will find.
- **Institute of Physics** — `https://www.iop.org` — search for material on vectors and their addition; it maps onto the vector subtraction of gravity.
- **Raspberry Shake** — `https://raspberryshake.org` — a global network of school seismographs with open data; pupils can compare their series with a real one.
- **Romanian General Inspectorate for Emergency Situations** — `https://www.igsu.ro` — for the natural continuation into *Earthquake!*

9. Assessment

The exit ticket, identical to the one on the pupil sheet: *what does the event counter actually count?* Collected in the last two minutes, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Explorer and Scientist level the three axes look like this:

- **Measurement practice** — the pupil notes the counter before and after each interval, keeps the direction constant within a set, does not move the box between days, and fills in the log.
- **Data interpretation** — the pupil uses the counter's rise as a count of shakes the box felt rather than of earthquakes, and compares intervals of equal length.

- **Scientific reasoning** — the pupil explains, through vector addition, why gravity has to be subtracted as a vector, and works out how much direction would have mattered if it had not been.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

10. Pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/seismology.pdf`. It has two pages — the direction test and the duration test on the first; the comparison between places or between days, and the conclusion, on the second — so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **Every table has "before / after / difference" columns** rather than a single value column: the counter is cumulative, and what matters is always its rise over an interval. **The direction column is compulsory and printed**, because direction is the activity's hidden variable — a mixed set means nothing. **The sheet explicitly asks for the length of each interval**, so that pupils compare only equal intervals — or divide the rise by the minutes when the lengths differ. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.